01

Present Value Fails

Prof. David C. Shimko

Industry Full Professor, NYU Tandon, Finance and Risk Engineering Dept.

Founder, Wall Street Scholars

September 2026

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Questions addressed here

What is PV and why is it important?

What are the discounting conventions used in PV calculations?

How do we value perpetuities?

Does PV provide sufficient information to make good financial decisions?

When should PV be used, and when should it not?

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What is PV and why is it important?

Suppose you are to receive a bonus at your company, but you have many choices as to how the bonus might be paid.

Alternatives include cash now, deferred cash, options, shares of the company, and different combinations thereof. Each combination may have a different pattern of future cash flows.

The idea of present value is to determine what the value of each of these patterns is today, after adjusting for timing and risk of payments. Choose the alternative with the highest present value!

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Discounting One Period

One period, risk-free

V_0: Present value (PV)

V_1: Future value (FV)

$V_1 - V_0$: Capital gain
(or increase in account value)

r: Annual risk-free return in percent

rV_0: Annual return in dollars
(time value of money)

Take note of the notation!
You will see it again…

One-Period PV Logic

Invest/deposit V_0 in the bank

Earn risk-free interest at annual rate r

Collect future value V_1 in one year

Then $V_1 = V_0 + rV_0 = (1+r)V_0$

OR

If you want to have V_1 available in one year, you must invest $V_0 = V_1/(1+r)$ now

Question:

What does this rearranged equation imply?

$rV_0 = (V_1 - V_0)$

Answer:

Time value (opportunity cost of cash) = Capital gain

The increase in the bank account value compensates you for the time value of investing (giving up or renting out) your money

What type of equation is this?

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Discounting Multiple Periods

"n" periods, risk-free

V_0: Present value

C_i: Riskless cash payment made at time i

V_n: Future value in n years

r: Annual return in percent

n-Period PV Logic

Invest/deposit V_0 in the bank

Earn risk-free interest at annual rate r

Collect future value V_n in n years

$V_1 = V_0 + rV_0 = (1+r)V_0$

$V_2 = V_1 + rV_1 = (1+r)^2 V_0$

* * *

$V_n = V_{n-1} + rV_{n-1} = (1+r)^n V_0$

OR

If you want to have V_n available in n years,

you must invest $V_0 = V_n/(1+r)^n$ now

Question:

What is the equation for the present value of n annual cash flows C_1, C_2, … C_n?

Answer:

$$V_0 = \sum_{i=1}^{n} \frac{C_i}{(1+r)^i}$$

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Continuous Discounting

Infinite number of discounting periods

k: Number of compounding periods per year

Simple interest definition: Divide simple annual interest rate r by k to determine the periodic interest rate

Continuous Compounding Logic

Invest V_0 in the bank

Earn risk-free interest at rate r/k per period, and let k become infinite

Compound k times per year (showing k=2, semiannual)

Then $V_{0.5} = V_0 + (r/2)V_0 = (1+r/2)V_0$

$V_1 = V_{0.5} + (r/2)V_{0.5} = (1+r/2)^2 V_0$

* * *

$V_n = V_{n-0.5} + (r/2)V_{n-0.5} = (1+r/2)^{2n} V_0$

Note as $k\ goes\ from\ 2\ to\ \infty$

(continuous compounding)

$$V_n \to V_0 e^{rn}$$

Therefore, to achieve value V_n at time n, if interest is compounded continuously at a simple annual rate of *r*, discount as follows to find the PV:

$$V_0 = V_n e^{-rn}$$

Questions:

What is the equation for the present value of n annual cash flows C_1, C_2, … C_n?

What if interest is compounded continuously on discrete payments?

What if payments are continuous? (Payment at t is $C_t dt$, ending at T)

Answers:

$$V_0 = \sum_{i=1}^{n} \frac{C_i}{(1+r)^i}$$

$$V_0 = \sum_{i=1}^{n} C_i e^{-ri}$$

$$V_0 = \int_0^T C_t e^{-rt} dt$$

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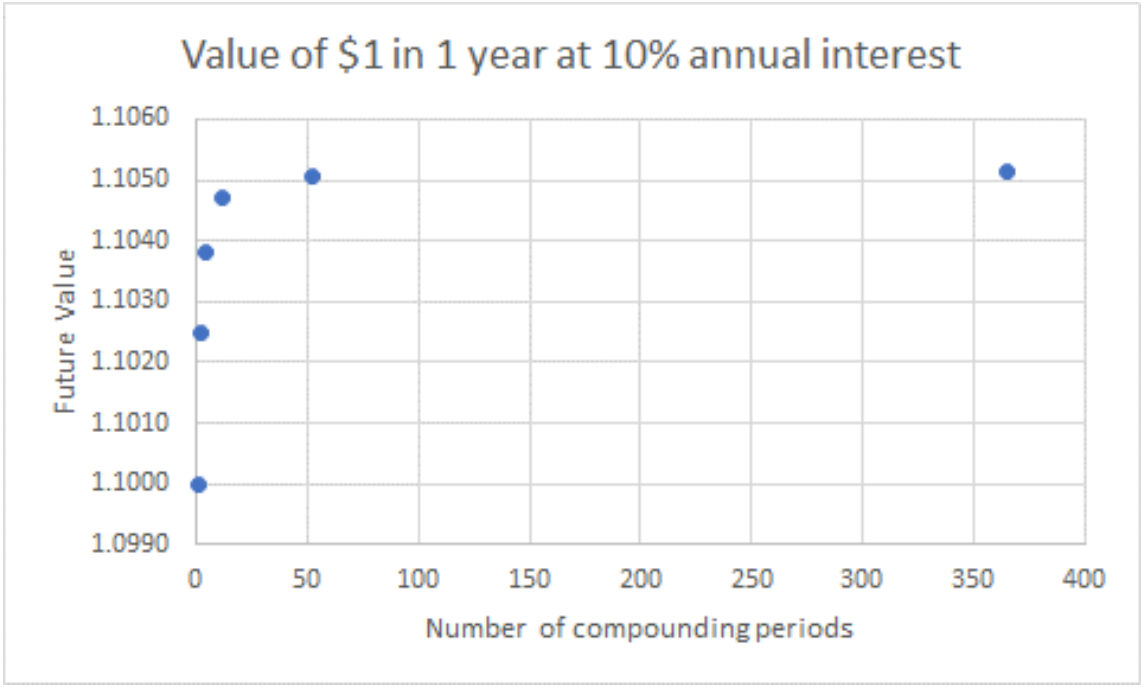
Yield based on compounding periods/yr

After 1 year, $1 invested at 10% will be worth...

...if compounded k times per year

k	FV	LN
1	1.1000	9.53%
2	1.1025	9.76%
4	1.1038	9.88%
12	1.1047	9.96%
52	1.1051	9.99%
365	1.1052	10.00%

Note: $\exp(r) = 1.1052 = e^{0.10}$



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Present Values can be easily computed in Excel™

So don't worry too much about the formulas!

	A	B	C
1	Discount rate		10%
2	Growth rate		20%
3			
4	Period	Cash Flow	
5	1	1,000.00	=NPV(C1,B5:B10)
6	2	1,200.00	$6,855.10
7	3	1,440.00	
8	4	1,728.00	
9	5	2,073.60	
10	6	2,488.32	



Application:

What is the present value of an infinite series of constant payments "C" starting one year from today, if the risk-free discount rate is r?

PV of a *perpetuity*

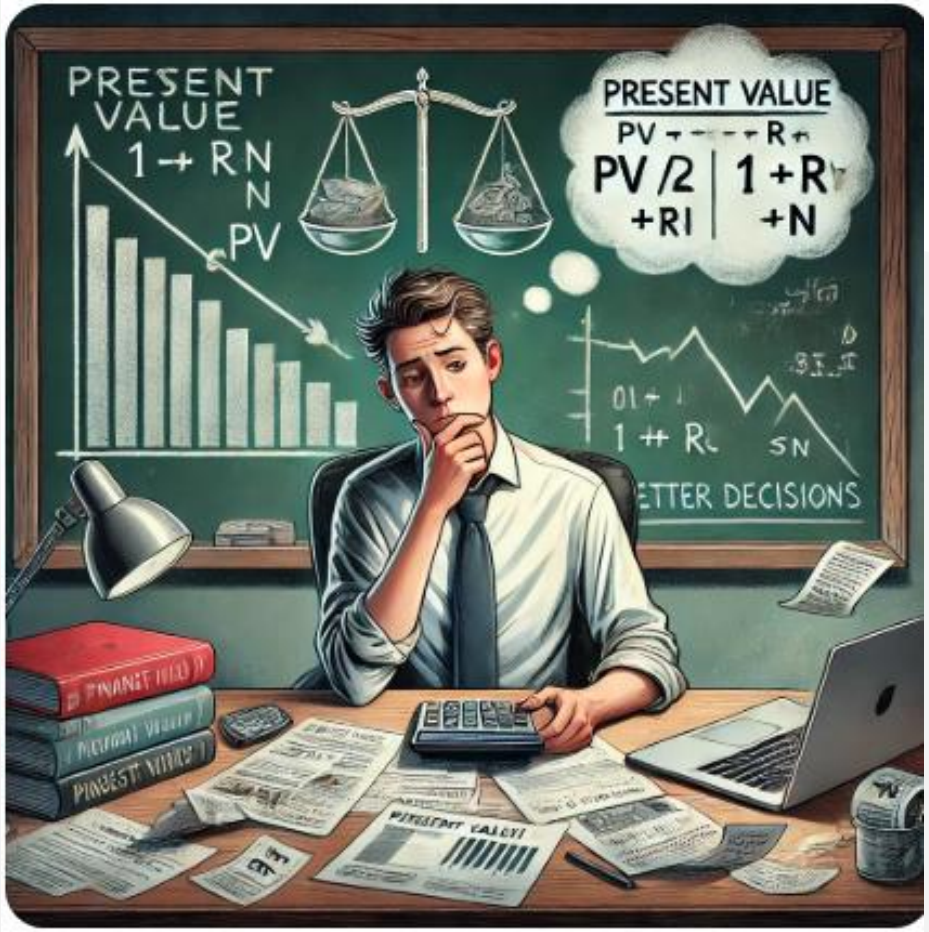
$$V_0 = \sum_{i=1}^{\infty} \frac{C}{(1+r)^i} = Cd \sum_{i=0}^{\infty} d^i \quad if\ d = \frac{1}{1+r}$$

Since d<1 this is a convergent geometric sequence

$$V_0 = Cd\frac{1}{1-d} = \frac{\frac{C}{1+r}}{1-\frac{1}{1+r}} = \frac{C}{(1+r)-1} = \frac{C}{r}$$

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Does *Present Value* provide the correct valuation for me?

And if not, when does it?

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Quiz 1 (risk-free)

Would you prefer $10,000 in one year or $11,000 in two years, assuming the interest rate is 5% p.a.?

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Actually, **I would prefer $10,000 in one year.** Am I wrong? Should I take the higher PV?



Present Values

$10{,}000/1.05 = \$9524$

$11{,}000/1.05^2 = \$9977$

How could she rationally prefer $10,000 in one year if the PV is lower? *Perhaps…*

- Cash constraints – Needs to buy a car
- Need to pay off high-interest debt
- Can invest the 10K to earn a return higher than 5% in year 2
- The obligor (payer) may default in the second year

Question

- **What could convince her to take the higher PV deal?**

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- **What could convince her to take the higher PV deal?**

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Answer

If the promised cash payments
can be sold now
at the stated discount rate
(to a bank, or the capital market),
then she will prefer to take $11K
in two years
if it has higher PV today.

What does this say about present value?

- Personal valuations generally differ from market values of assets
- PV implicitly assumes cash flows are marketable (or that they may be treated as such), i.e. they can be sold today
- PV analysis should always restate this implicit assumption
- **PV is only one model of valuation, not a foolproof solution**



Quiz 2

The state of California requires drivers to either purchase auto insurance or post a $30,000 bond with the state. You may keep the 5% interest paid on the bond, but if there is a loss, the bond is liquidated to cover damages.

You have estimated your expected loss on auto insurance to be $2500 per year, because you are a good driver. You also expect your loss to be $1000 per year if you do not buy insurance.

Which types of people will prefer to buy insurance, and which will prefer to post a bond?

Assume:

- One year only
- Maximum liability in accident = $30,000
- You don't care about risk, only expected loss

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Solution

- A recent college graduate without $30,000 has no choice but to buy insurance in CA
- For someone with the cash, it appears that the PV savings of self-insurance is $1500 per year, so it is preferable to post the bond
- However, if one can invest $30,000 in a bond, then they should buy insurance if their foregone return on capital for buying the bond is greater than the expected loss due to paying for overpriced insurance. This will be true if their opportunity cost of capital exceeds 10%.

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Teachings on PV

1. PV provides a way to compare cash flow streams with different payment amounts and timing.
2. PV rewards acceleration of positive cash flows and deferral of negative cash flows.
3. PV of risk-free assets has nice mathematical properties: Additive values, scalable and separable. The sign of the PV is the same as the sign of the cash flows, as long as all cash flows themselves have the same sign. **These properties do not generally apply when cash flows are risky.**
4. Compounding conventions are important. Corporate analysis requires discrete discounting, while derivatives (i.e. futures and options) require continuous discounting. The conversion of simple annual rates to periodic rates is critical for accurate bond analysis.
5. PV does not apply automatically to non-marketable cash flows, we must make this assumption explicitly in order to use PV analysis.
6. Above all else, PV is but one model of value. It is subject to assumptions that need to be tested, and may even be false.

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Potential PV Fails

1. Dependence on Discount Rate
The choice of discount rate has an outsized effect on the result.
Small changes in the rate can lead to large swings in valuation, making PV highly sensitive and sometimes arbitrary. It's often unclear whether to use a risk-free rate, weighted average cost of capital (WACC), or some other rate—introducing subjectivity.

2. Forecasting Problems
PV relies on projecting future cash flows, but those forecasts are inherently uncertain.
For long-lived projects (e.g., infrastructure, R&D, or startups), small errors in assumptions about growth, margins, or terminal values compound drastically. It implicitly assumes the forecaster can model an uncertain future with precision, which is rarely true.

3. Linear, Deterministic Thinking
PV assumes cash flows occur in a predictable, linear fashion.
It does not capture real options value, flexibility, or path dependency (e.g., management's ability to adapt strategy if conditions change). As a result, it undervalues projects where adaptability is a key source of value.

4. Time Value Overdominance
PV heavily discounts distant cash flows.
This can make long-term benefits appear negligible, even if they are large in absolute terms. As a result, it biases decision-making toward short-term projects and against sustainability or strategic initiatives with long payback horizons.

5. Risk Misrepresentation
PV typically addresses risk by adjusting the discount rate, but this is a blunt instrument.
Different risks (systematic vs. idiosyncratic, diversifiable vs. non-diversifiable) are not differentiated. It ignores the distribution of outcomes and only produces a single-point estimate.

6. Ignores Market Dynamics
PV is not forward-looking in the same way markets are.
It assumes investors act on static projections rather than continuously updating based on new information, competition, and strategic positioning.
Thus, it can diverge from actual market valuation.

7. Terminal Value Issues
In discounted cash flow (DCF) models, the terminal value often represents the majority of PV.
Terminal value relies on arbitrary perpetual growth rates or exit multiples, which are hard to justify and distort the analysis.
This creates a false sense of precision while actually being highly speculative.

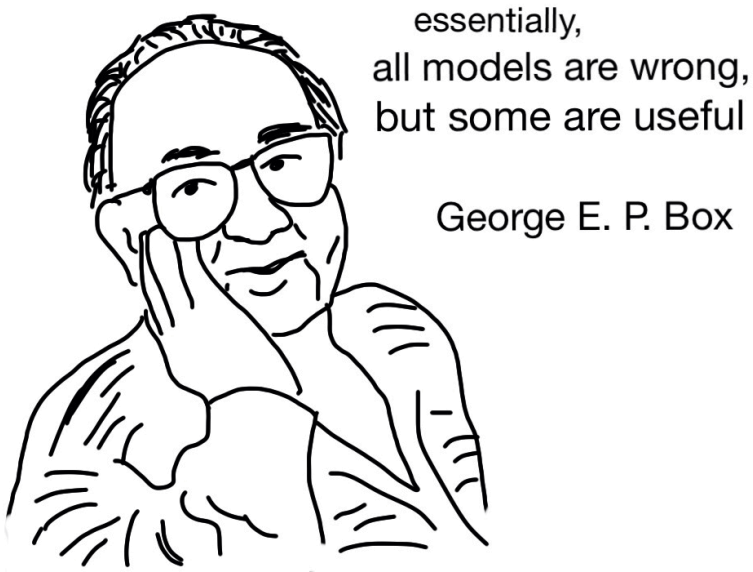
8. Ethical and Social Blind Spots
PV only considers financial returns, ignoring externalities such as environmental or social impacts.
For public policy or ESG-oriented investment, this is a major failing.
It implicitly assumes value is only private and monetary, not societal.

9. No Incorporation of Uncertainty/Optionality
PV does not naturally account for probabilities of failure or upside scenarios.
Real options theory, decision trees, and probabilistic methods do a better job here.
PV assumes "average" projections, but in uncertain environments, averages can be misleading.

👉 In short: **Present Value fails as a model because it is overly sensitive, highly assumption-dependent, and reductionist—it compresses uncertainty, adaptability, and strategic considerations into a single number that often hides more than it reveals.**

Last Words

1. Present value (PV) is only one model of valuation, not the definitive model.
2. All models are wrong, because we make simplifying assumptions that are wrong.
3. PV can be only be considered reliable in situations where the cash flows can be sold today at fair value.
4. Otherwise, we may need better valuation models.



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